Assignment 2. Gauss-Jordan & LU Marks 20

Posted on 19.08.2026 @ 2:30 pm and due on 19.08.2026 @ 5:40 pm

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1. Prepare library routines for Gauss-Jordan elimination method using *row pivoting* and LU decomposition. For LU mention your choice of *Doolittle* or *Crout* clearly. **[5+5]**

2. Check Gauss-Jordon with the system of equations for three variables. **[5]**

$$\begin{aligned} 2y + 5z &= 1 \\ 3x - y + 2z &= -2 \\ x - y + 3z &= 3 \end{aligned}$$

3. Find LU decomposition of the matrix and verify. **[5]**

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{pmatrix}$$